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CHEQ-ing the Box: Safe Variable Impedance Learning for Robotic Polishing ⋆

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Abstract: Robotic systems are increasingly employed for industrial automation, with contact-rich tasks like polishing requiring dexterity and compliant behavior. These tasks are difficult to model, making classical control challenging. Deep reinforcement learning (RL) offers a promising solution by enabling the learning of models and control policies directly from data. However, its application to real-world problems is limited by data inefficiency and unsafe exploration. Adaptive hybrid RL methods blend classical control and RL adaptively, combining the strengths of both: structure from control and learning from RL. This has led to improvements in data efficiency and exploration safety. However, their potential for hardware applications remains underexplored, with no evaluations on physical systems to date. Such evaluations are critical to fully assess the practicality and effectiveness of these methods in real-world settings. This work presents an experimental demonstration of the hybrid RL algorithm CHEQ for robotic polishing with variable impedance, a task requiring precise force and velocity tracking. In simulation, we show that variable impedance enhances polishing performance. We compare standalone RL with adaptive hybrid RL, demonstrating that CHEQ achieves effective learning while adhering to safety constraints. On hardware, CHEQ achieves effective polishing behavior, requiring only eight hours of training and incurring just five failures. These results highlight the potential of adaptive hybrid RL for real-world, contact-rich tasks trained directly on hardware.

Keywords: reinforcement learning, variable impedance, robotic polishing

1. INTRODUCTION

Robotics has become a cornerstone of industrial automation, playing a pivotal role in improving efficiency and precision across various domains. A long-term aspiration in robotics is to achieve manipulation capabilities that combine human-like dexterity with compliant behavior, enabling robots for contact-rich tasks. In recent years, learning-based approaches, such as deep reinforcement learning (RL), have emerged to learn complex nonlinear control policies directly from data. While RL has shown great success in challenging control problems such as gameplay (Mnih et al., 2015; Silver et al., 2018a) and robotic manipulation (Büchler et al., 2022), its application to real-world problems remains limited mainly due to its data inefficiency and unstructured exploration behavior. Prior work often focuses on sim-to-real transfer to mitigate these challenges. However, for highly complex or contact-rich tasks, it is often impractical to design accurate simulations that capture all necessary dynamics and interactions. As a result, direct learning on hardware sometimes becomes unavoidable. This introduces a critical challenge: the reliance of RL on random exploration can lead to unsafe behaviors, risking damage to both the robot and its environment. Addressing safe exploration is essential to enable safe and efficient RL training on hardware.

A prime reason for the random exploration behavior is the task-agnostic architecture of state-of-the-art RL approaches (Haarnoja et al., 2018), which lack the incorporation of prior knowledge on how to solve the task at hand. In contrast, control theory offers a rich set of methods for deriving near-optimal controllers based on first-principles modeling. These methods provide a decent baseline that works well when the system dynamics can be accurately modeled but fall short when the dynamics are complex or poorly understood. This motivates hybrid RL (Silver et al., 2018b; Johannink et al., 2019), blending control priors with deep RL policies. Hybrid algorithms thus combine the informed behavior of the prior with task-specific RL optimization to tackle complex, nonlinear problems.

The majority of prior work in hybrid RL (Silver et al., 2018b; Johannink et al., 2019; Schoettler et al., 2020; Ceola et al., 2024) proposes a fixed weighting between control prior and RL agent. A fixed blending, however, disregards that the capability of the RL agent depends on training time and state. As more data is collected, the agent refines its behavior, ultimately outperforming the prior across larger portions of the task domain. Adaptive hybrid RL (AHRL) methods (Cheng et al., 2019; Rana et al., 2023) adapt the weighting between RL agent and control prior based on the agent's confidence. The newly developed CHEQ algorithm (Cramer et al., 2024a) dynamically adjusts the weighting based on the parametric uncertainty of a critic ensemble. This has demonstrated significantly safer exploration behavior, i.e., fewer violations of safety limits,

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and accelerated learning than traditional model-free RL, residual RL, and prior work in AHRL. The properties of AHRL methods, in general, and CHEQ, in particular, are promising for real-world applications and direct training on hardware. Hardware evaluations are critical, as real-world challenges like contact dynamics, friction, and state-dependent noise are difficult to simulate accurately. While some fixed-weight hybrid RL methods have trained on hardware (Johannink et al., 2019; Schoettler et al., 2020), the advancements of AHRL have been validated exclusively in simulation and with low-dimensional action spaces. We argue that the full potential and the true challenges of these algorithms can only be revealed through evaluation on a challenging hardware problem.

In this work, we focus on polishing a 3D object with a robotic arm, a task that exemplifies the challenges of contact-rich manipulation in industrial automation. Polishing requires continuous contact with a surface while adhering to specific force and velocity profiles, demanding precise control to ensure quality and consistency. Variable impedance control (VIC) is particularly well-suited for such tasks (Martín-Martín et al., 2019; Bogdanovic et al., 2020), as it enables robots to dynamically adapt their stiffness and damping to interact effectively with their environment. However, defining suitable variable impedance gains is inherently complex, and manual tuning becomes impractical in scenarios with high-dimensional or nonlinear dynamics. This makes robotic polishing a compelling benchmark for evaluating RL-based control strategies.

This work is the first to apply AHRL, specifically the CHEQ algorithm, to a hardware learning problem. The hybrid RL agent outputs the impedance gains and the end-effector position and orientation. This output is then fed into a cascaded impedance controller. The goal is to (i) investigate the potential of CHEQ for learning variable impedance gains and (ii) explore its potential to learn such a policy safely, directly on hardware.

In simulation, we demonstrate that VIC significantly improves polishing behavior. We compare standard RL with AHRL, showing that CHEQ can learn effective impedance gains while maintaining exploration within safety limits. While CHEQ cannot provide theoretical safety guarantees, the control prior guides the exploration behavior to be close to the desired task. Finally, we deploy CHEQ on hardware and show that good polishing behavior can be achieved within eight hours and only five failures, underscoring its efficiency and safety in real-world scenarios.

2. RELATED WORK

In this section, we discuss relevant prior work in hybrid RL and works focusing on RL-based VIC and polishing.

Hybrid Reinforcement Learning. Hybrid RL combines RL and control priors, categorized into fixed or adaptive weighting.

Silver et al. (2018b); Johannink et al. (2019) first combined RL and control, introducing the term residual RL. In this work, we use the more general term hybrid RL to include approaches that adapt the controller's weight. Silver et al. (2018b) and Johannink et al. (2019) show advantages of hybrid RL, such as sample efficiency and improved sim-to-real transfer. Fixed weight hybrid RL has been applied to tasks such as real robot insertion tasks (Schoettler et al., 2020; Davchev et al., 2022) and residual grasping policies on top of a simulation trained policy (Ceola et al., 2024). A fixed mixing, however, does not consider the improving capabilities of the RL agent.

In contrast, AHRL adjusts weighting based on RL agent capabilities. Rana et al. (2023) use a policy ensemble to estimate how certain the RL agent is in the current action. The combined action reflects the Bayesian posterior of control prior and policy distribution. Cheng et al. (2019) use the TD-error as an uncertainty estimate and combine controller and RL agent based on this. Rana et al. (2023) and Cheng et al. (2019) train based on the combined action, which becomes brittle when facing large distributional shifts. Cramer et al. (2024a) demonstrate that the weighting factor induces a contextualized MDP, which must be observable by the agent. They train based on the RL action, incorporating the weighting factor into the agent's state. This approach achieves significantly safer exploration and fewer failures than standard RL, fixed weighting, and prior adaptive methods. This makes it an ideal AHRL method to investigate safe exploration during training on real hardware, which we do herein.

Reinforcement Learning-based VIC and Polishing. Reinforcement Learning can be used to learn VIC, first demonstrated by Buchli et al. (2011). Some works have focused on learning VIC from data gathered on hardware. Luo et al. (2019) combine a higher model-based RL controller with a lower-level impedance controller, showing that this approach can learn real-world insertion tasks data efficiently. In (Roveda et al., 2020), a model-based RL approach is trained to model human-robot interaction dynamics, and then an MPC is used to optimize VIC parameters. Anand et al. (2024) learn VIC with a model-based RL agent through real-world interactions. They evaluate their approach on water-pouring with a robotic arm. While prior work emphasizes data-efficient model-based RL, both model-based and model-free algorithms are commonly used in RL. Here, we use model-free hybrid RL to remain safe during exploration.

Some prior work specifically focused on learning VIC for robotic polishing. Martín-Martín et al. (2019) investigated vision-based surface wiping. While we aim to track a specific target force, they want to achieve an arbitrary force sufficient to wipe dirt. They trained in simulation and then transferred the policy to a real robot. Bogdanovic et al. (2020) examined the simulated task of a circular end-effector motion while in contact with a table and applying a desired constant vertical force. Anand et al. (2022) consider wiping a 2D surface with a robotic arm, demanding force tracking in the vertical direction and motion tracking in the remaining five DOFs. The model-based RL agent tunes the two-dimensional impedance gains, while motion tracking is handled by a cascaded controller. This approach was directly trained on hardware, focusing on data efficiency. All approaches above consider wiping a 2D surface, which substantially reduces the action space of the RL agent. In this work, we want to use a more complex 3D task that requires adapting impedance gains for all

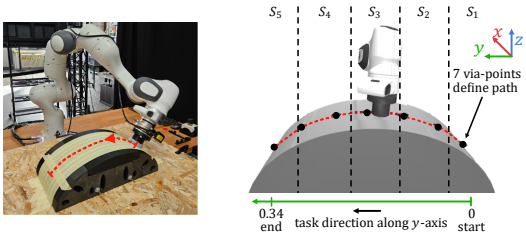

Fig. 1. Real robot setup and task description.

Cartesian axes and orientations. Both Martín-Martín et al. (2019) and Bogdanovic et al. (2020) investigate different choices of action spaces and conclude that a cascaded RL approach with variable impedance in end-effector space is most suited for the considered wiping task. We thus use this approach as our control pipeline and investigate the potential of AHRL for training directly on hardware.

3. ROBOTIC POLISHING

We explore robotic polishing as a continuous contact force control task solvable with VIC. A non-rotating polishing tool is attached to a 7-DOF Franka Panda robotic arm, with a 3D-printed bridge as workpiece. We choose this object for its multiple challenges. The 3D movement showcases the need for variable impedance. Further, the task's movement direction does not align with Cartesian axes, and the workpiece features curvature variation along the y-axis, adding complexity.

The task involves polishing along a path r^{path} defined via 7 distinct via-points. Figure 1 shows the simulated and real robot setup and the desired polishing path. This research aims to achieve fine material removal through precise, low-force polishing, maintaining consistent material removal. Assuming a constant contact area, the material removal rate (Fiedler, 1998) is $\frac{dh}{dt} = c \cdot F_N \cdot v_T$, with material thickness h, normal force on the workpiece F_N, translational velocity v_T along r^{path}, and c a proportionality constant. Consistency in material removal rate thus requires maintaining constant force and translational velocity throughout the task.

4. LEARNING VARIABLE IMPEDANCE VIA CHEQ

Our control pipeline (cf. Fig. 2) comprises a hybrid RL-based reference controller (RC) and cascaded impedance controller in end-effector space (IC) (Khatib, 1995). The RC is based on our AHRL algorithm, CHEQ. This controller outputs a_t^{ref}, including the next Cartesian end-effector pose and variable impedance gains. The IC outputs $a_{\hat{t}}^{\text{IC}}$, mapping end-effector pose error to desired joint torques while keeping compliant end-effector behavior.

4.1 Impedance Controller (IC)

We specify the robot end-effector pose $p \in \mathbb{R}^6$ by its position $x = (x_x, x_y, x_z)$ in Cartesian space and orientation $\phi = (\phi_x, \phi_y, \phi_z)$ in Euler angles. The complete end-effector pose can be written as $p = (x, \phi)$. We denote the pose error between the target and the current pose at time t as $\Delta p = p^{\text{ref}} - p$. End-effector velocity and acceleration are defined as $\dot{p}$ and $\ddot{p}$ and their errors as $\Delta\dot{p}$ and $\Delta\ddot{p}$ respectively. We denote the joint space configuration by $q \in \mathbb{R}^7$. The goal of impedance control is to model the behavior of a mass-spring-damper system in task space, such that $M\Delta\ddot{p} + D\Delta\dot{p} + K\Delta p = f_{\text{ext}}$. Here M, D, and $K \in \mathbb{R}^{6\times 6}$ represent the Cartesian inertia, damping, and stiffness matrices, and f_{ext} is a force applied to the robot. Commonly, $M = 0$ is assumed to mitigate noisy acceleration measurements (Lynch and Park, 2017). Moreover, we set $\dot{p}^{\text{ref}} = 0$. Under these assumptions, the desired joint torques to achieve this impedance behavior can be calculated as $\tau_{\text{IC}}(q) = J^{\intercal}(K\Delta p - D\Delta\dot{p})$, where J is the Jacobian matrix. The controller also applies a null-space torque $\tau_{\text{null}}(q)$, Coriolis and centrifugal forces $C(q, \dot{q})$, and gravity compensation $g(q)$. The IC then outputs $a_{\hat{t}}^{\text{IC}} = \tau_{\text{IC}}(q_{\hat{t}}) + \tau_{\text{null}}(q_{\hat{t}}) + C(q_{\hat{t}}, \dot{q}_{\hat{t}}) + g(q_{\hat{t}})$.

4.2 Hybrid RL-based Reference Controller (RC)

We use the CHEQ algorithm (Cramer et al., 2024a) as the RC, adaptively combining actions from the RL agent and nominal controller.

RL Agent. We model the environment as a discounted Markov decision process defined by the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, p, r, \rho_0, \gamma)$, with state space $\mathcal{S}$, action space $\mathcal{A}$, start state distribution ρ_0 and transition function $p(s_{t+1}^{\text{RL}}, r_{t+1} \mid s_t^{\text{RL}}, a_t^{\text{RL}})$. During transitions, rewards $r_t \in \mathbb{R}$ are emitted. The objective is to learn a policy $\pi^{\text{RL}}(a_t^{\text{RL}} \mid s_t^{\text{RL}})$ that maximizes the expected cumulative sum of rewards discounted by $\gamma \in (0, 1)$. This objective can be written as $J(\pi^{\text{RL}}) = \max_{\pi^{\text{RL}}} \mathbb{E}_{\pi^{\text{RL}}, \mathcal{M}} [\sum_{t=0}^{\infty} \gamma^t r_{t+1}]$. The action value or Q-function $Q^{\pi^{\text{RL}}}(s_t^{\text{RL}}, a_t^{\text{RL}})$ conditions expected return on specific state action pairs.

CHEQ builds upon the popular model-free soft-actor critic (SAC) architecture (Haarnoja et al., 2018) and combines this with an ensemble of E critic networks. To reduce the action space, we define the stiffness matrix as a diagonal matrix $K = \text{diag}(k_x, k_y, k_z, k_{\phi_x}, k_{\phi_y}, k_{\phi_z})$ and the damping by a damping factor ζ proportional to K such that $D = 2\zeta \cdot \sqrt{K}$. The agent then outputs

$$a_t^{\text{RL}} = (\Delta p_t^{\text{RL}}, K_t, \zeta_t).$$

For the state space of the agent, we define $\tilde{p} = (x, \phi^{\text{quat}})$ where $\phi^{\text{quat}} \in \mathbb{R}^4$ describes the orientation in unit quaternions. The complete state space can then be written as

$$s_t^{\text{RL}} = (q_t, \cos(q_t), \sin(q_t), \dot{q}_t, \tilde{p}_t, \dot{\tilde{p}}_t, F_t, \Delta r_t^{\text{path}}) \in \mathbb{R}^{74}.$$

Here, $F_t \in \mathbb{R}^3$ is the contact force of the end-effector, and $\Delta r_t^{\text{path}} \in \mathbb{R}^{5\times 6}$ describes the position and velocity errors of the end-effector to the next five points along the path r^{path}. The contact force is sensed via a force-torque sensor.

At each time step t, the reward is a weighted sum of terms designed to encourage path-following, as well as velocity and force tracking such that

$$r_t = c_{c^\perp} r_{c^\perp,t} + c_{c^\parallel} r_{c^\parallel,t} + c_d r_{d,t} + c_v r_{v,t} + c_f r_{f,t}.$$

Here, $r_{c^\perp}$ and $r_{c^\parallel}$ reward path-following in perpendicular and parallel path direction, r_d directional alignment, and r_v and r_f velocities and forces close to the respective targets. Each term is weighted with a constant c. In addition, we give a penalty r_{trunc} for episode truncation and a final termination reward r_{term} when all via-points are successfully wiped. Details are in Cramer et al. (2024b).